BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 300 · Heuristics, modelling and pipelining

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a heuristic provide?

Q2. What is throughput?

Q3. What is a pipeline stall?

Q4. Binary search for 13 in indexes0–14 of [0,1,...,14]. List midpoint indexes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

heuristic — Estimate/rule guiding search.

performance model — Simplification estimating resource use.

pipeline — Overlap processing stages.

stall — Delay in progress through stages.

New learning

Define the heuristic — A heuristic quickly estimates how promising a choice is; it is not automatically an exact solution.

Model the trade-off — Separate local edge cost, accumulated cost and estimated remaining cost before choosing a path.

Pipeline and performance models — A pipeline can overlap independent stages, increasing throughput, but dependencies and resource constraints may cause stalls.

Worked example

Evaluate two routes A–B–G (3+7) and A–C–G (5+1).

1. Predict first — Calculate complete route costs, rather than deciding only from the first edge.

2. Explain the method — The via-C path is cheaper at six total units despite a larger first edge.

We Do · Practise with guidance

A three-stage pipeline receives four independent tasks, one task per stage per cycle. When does the fourth finish?

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Stages S1,S2,S3
Task4 starts S1 in cycle4
Finishes S3 in cycle6
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Predict first

Task one occupies S1 in cycle 1, with one new task entering each later cycle.

Explain the method

Task four completes stage S3 in cycle 6, assuming no stalls; it is not completed in cycle 4.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain why choosing the cheapest immediate edge is not guaranteed to produce a shortest path. [4]

Question 2

Five independent jobs pass through 3 one-cycle stages. Give ideal completion cycle for all five and compare with non-overlapped 15 cycles. [6]

Question 3

Evaluate using an A* heuristic and a computational performance model to plan emergency routes. [6]

Exit Ticket · Answer before checking

Choosing the cheapest immediate edge always yields the cheapest complete path.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A rule-of-thumb estimate useful for choosing or ordering options.

Q2. Completed work per unit time.

Q3. Delay caused by unavailable stage/result or resource.

Q4. mid7, then mid11, then mid13; found at 13.

You Do pages · Independent questions

Q1. Local choice ignores later costs [2]; example via B 2+8=10 vs C 5+1=6 [2].

Q2. One starts each cycle [2], pipeline fill3 [1], last completes cycle3+5−1=7 [2], non-overlap=15 [1].

Q3. Heuristic can direct search [1], must not overestimate for optimality under appropriate assumptions [1]; live network weights may change [1]; model validation/data quality [1], congestion/resources [1], compare quality and computation time [1].

Exit Ticket · Model answer

False. Later costs can reverse the choice.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.